

PHYSICS, CHEMISTRY & MATHEMATICS

QP Code:

Test - 6

Time Allotted: 3 Hours

Maximum Marks: 180

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. In Each Section is **One Part**: Part-A.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with HB pencil for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Sections.

- (i) **Part-A (01-15)** contains (15) Numerical based questions, the answer of which maybe positive or negative numbers or decimals (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) and each question carries **+4 marks** for correct answer and **there will be no negative marking**.

Name of the Candidate : _____

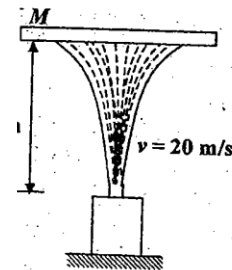
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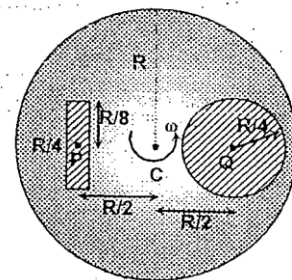
SECTION-1 : PHYSICS**PART – A**
(Numerical based)

This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

1. A vertical jet of water coming out of a nozzle with velocity 20 m/s supports a plate of mass M at a height $h = 10\text{m}$, as shown in the figure. If the rate of water flow is 1 litre per second then find the mass of the plate in kg. (assume the collision to be inelastic)



2. A rod of mass m and length $R/4$ and a disc of mass m and radius $R/4$ are hinged on larger disc of same mass and radius R . The rod and disc can freely rotate about hinge points P and Q . If the moment of inertia of the system about an axis which is passing through C and perpendicular to the plane of the larger disc is $k m R^2$ find the value of k



3. Moment of Inertia of the given isosceles triangular plate in figure-1, about an axis passing through O and perpendicular to plate I . If the moment of Inertia of the given plate is figure-2, about an axis passing through C and perpendicular to the plate as shown in the figure is kI then find k . (mass per unit area of both the plate is same).

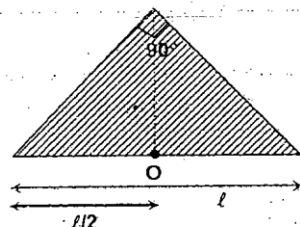


Figure -1

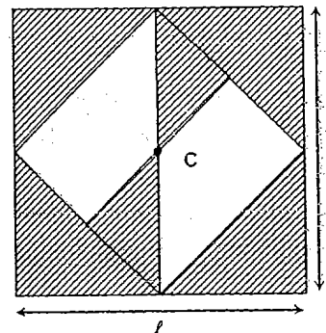
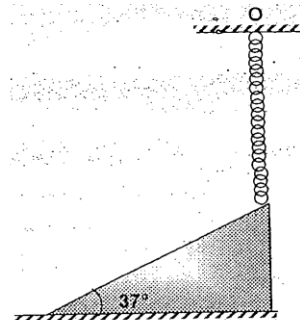
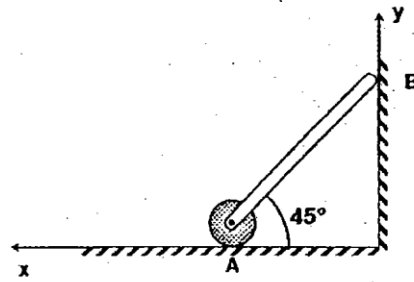


Figure -2

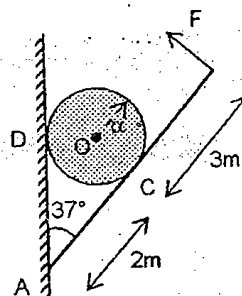
4. A thin flexible uniform chain of mass m and length l is suspended from point O so that its lower end just touches a smooth inelastic fixed plane of inclination 37° . Now the chain is released from point O then find the total Impulse on the plane while the complete chain falls is $k m \sqrt{g l}$ then find k .



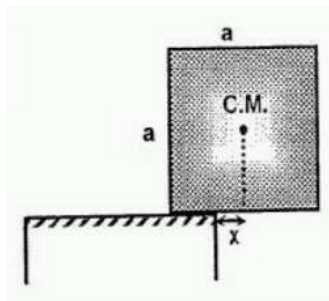
5. The uniform rod AB of mass 1 kg is supported on a horizontal smooth surface by a small roller of negligible mass and dimension. If the coefficient of kinetic friction between end and vertical wall is $1/3$. The rod is released from rest in the shown position then find the magnitude of initial acceleration of rod AB.



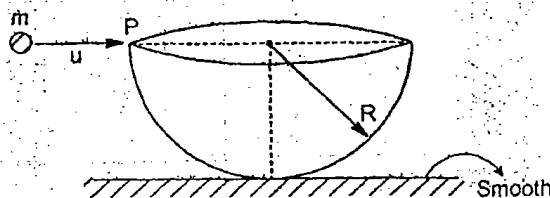
6. A cylinder of radius r and mass m is wedged between a vertical wall and a light hinged (at point) bar as shown in the figure. The coefficient of friction between cylinder and wall is 0.2 and between bar and cylinder is 0.4. Then find the minimum force which is just sufficient to cause the cylinder to slip upward, when applied at the end of the bar and at right angles to the bar. (take $m = 1\text{ kg}$, $g = 10\text{ m/s}^2$).



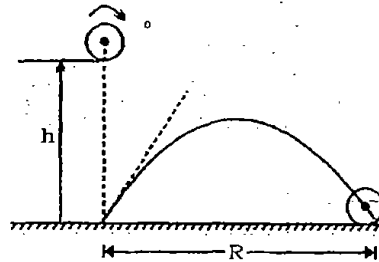
7. The uniform cubical slab of side a is released from rest in the position shown in the figure. For what value of x is the angular acceleration maximum (friction between the surface and the cube is sufficient to prevent any slipping). Take $a = 6\text{ m}$.



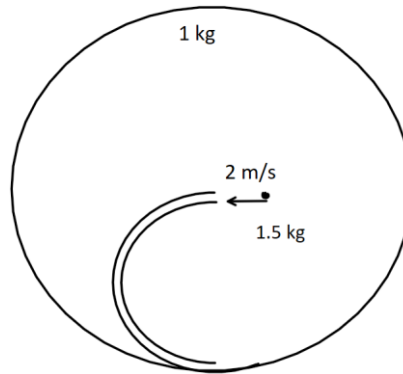
8. A hollow hemispherical bowl of mass m , radius R is placed on a smooth horizontal surface. A particle (also of mass m) strikes to it with speed u as shown in figure. The velocity of particle just after the collision in vertically upward direction is ku find k .



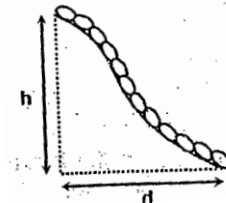
9. Figure shows a solid homogeneous ball of radius R . Before falling, the velocity of mass is zero, but the ball is rotating with angular velocity ω_0 about horizontal axis passing through the center. The ball is released, it collides with the floor (coefficient of restitution is e). The coefficient of friction between the ball and the floor is μ_k due to which the ball moves in a projectile. Assume that there is slipping throughout the impact and neglect air resistance. The horizontal range of the ball is given by kh . (take $e=1/2$, $\mu_k=1/12$) find k .



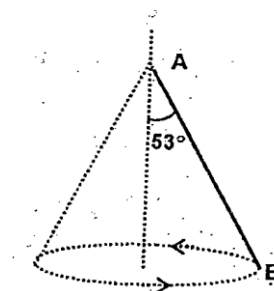
10. A disc of mass 1 kg, hinged at its centre, is free to rotate about its centre. A small disc of mass 1.5 kg moves with speed 2 m/s relative to ground and enters a tunnel formed on disc. Suppose formation of tunnel does not make any effect on the mass of disc. Find the velocity with which small disc come out of disc.



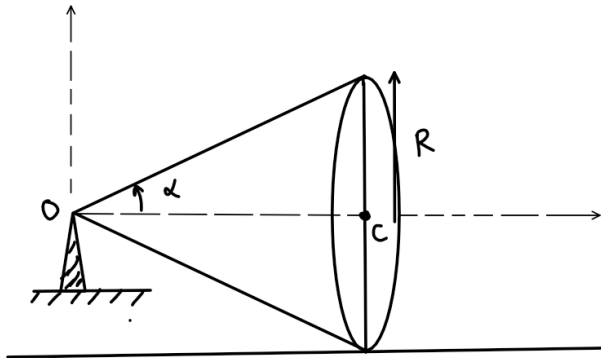
11. A uniform chain of mass per unit length λ is sliding down the hill with constant speed u . Dimensions of hill are h and d as shown. Find the momentum of chain at the instant shown. (given $\lambda = 1/4$ kg/m, $d = 3$ m, $h = 4$ m, $u = 5$ m/s)



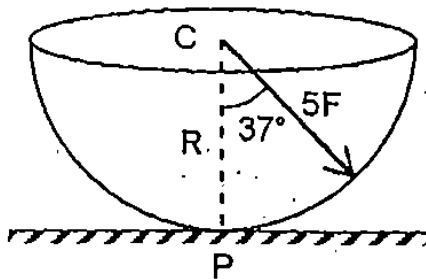
12. A uniform rod of length 4 m and mass 2 kg revolves with constant angular velocity ω about a vertical axis through a smooth joint A at one extremity of the rod so that it describes a cone of semi vertical angle 53° as shown in the figure. Find the magnitude of hinge reaction at point A.



13. A cone rolls without slipping on the horizontal surface so that its apex remains fixed and the axis of cone is horizontal. The radius of the base of the cone is r and the semi-vertex angle is α . The centre of base of the cone is C and it moves with constant speed v_0 . Find the total Kinetic energy of the cone. (take $m = 2\text{kg}$, $v_0 = 5\text{ m/s}$)



14. A half section of a uniform hollow sphere of mass m is at rest. A force $5F$ is applied from inner side of sphere as shown in figure. Assume that the section rolls without slipping. Find the minimum values of μ_s compatible for the motion.



15. Ball A is in motion with linear momentum P while ball B is at rest. Masses of both the balls are same and equal to m . Ball A is obliquely collide with the ball B. After the collision, ball A moves at an angle of 30 degrees with the initial direction while momentum of ball B is $\frac{2}{\sqrt{3}}$ times of its minimum possible value. Then find loss in kinetic energy after the collision. ($P = 3\text{ kg-m/s}$, $m = 1\text{ kg}$)

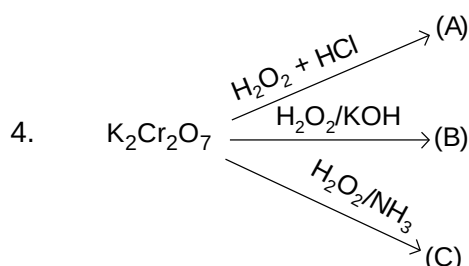
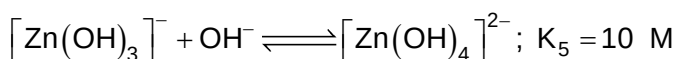
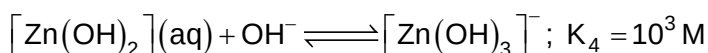
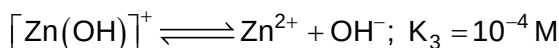
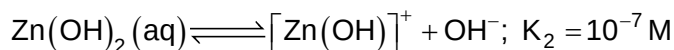
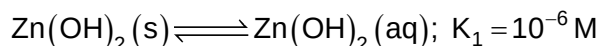
SECTION-2 : CHEMISTRY**PART – A**
(Numerical based)

This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

- How many of the given statement(s) is/are correct?
 - Metallic bond strength in 's'-block elements decreases down the group.
 - Metallic character increases down the group in 's'-block.
 - Reactivity of 's'-block elements increases down the group.
 - Solution of alkali metal in liquid NH_3 is blue in colour due to presence of solvated electron.
 - The blue colour of alkali metals in liquid NH_3 slowly fades until it disappears owing to the formation of a metal.
 - At concentrations about 3M, solution of alkali metal in liquid NH_3 are copper-bronze coloured and have a metallic lustre.
 - Conduction of alkali metal in liquid NH_3 is mainly due to the presence of solvated electron.
 - Solutions of alkali metals in liquid NH_3 act as a powerful reducing agent and they will even reduce an aromatic ring

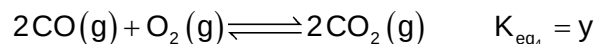
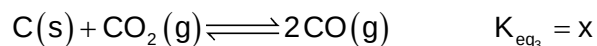
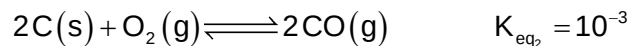
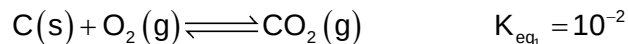
- How many of the given molecules have central atom sp^3 hybridized
 SF_4 , XeO_2F_2 , XeOF_2 , XeOF_4 , XeO_4 , CO_2 , SiO_2 , ClF_3 , CH_3^- , CH_3^+ , Carbonsuboxide

- If the solubility of solid zinc hydroxide at $\text{pH} = 5$ is x . What is the value of $x/2$?
 Given:



The oxidation number of Cr in product A is x , in product B is y and in product C is z . Value of $\frac{x+y}{z} =$

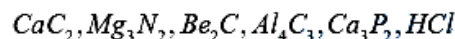
- The London force is proportional to r^{-x} , (where r is inter atomic/molecular distance). What is the value of x .
- At high temperature, the following equilibrium exist in a mixture of carbon, oxygen and their compounds



What will be the ratio of x and y?

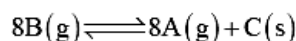
7.

How many of the following compounds produce a product having molecular weight equal to heavy water (D_2O) when dissolved in heavy water



8

A vessel contains A(g) and B(g) at 2 atm and 4 atm respectively at TK, the mixture is allowed to attain equilibrium at TK, according to the reaction



At equilibrium, $\left(\frac{n_A}{n_B}\right)_{\text{eq}} = \left(\frac{n_B}{n_A}\right)_{\text{initial}}$

Find the value of y if $K_c = 2^y$

9. What is the pH of 0.1 M of aq Borax solution (K_a for $H_3BO_3 = 2 \times 10^{-10}$) ($\log 2 = 0.30$)

10. The molarity of perhydrol solution is nearly equal to

11. 1 mole of borax reacts with y mole of HCl. y is

12. Calculate the pH after mixing 1 mole each of H_3PO_4 , NaH_2PO_4 , Na_2HPO_4 and Na_3PO_4

For H_3PO_4 $K_{a_1} = 10^{-3}$, $K_{a_2} = 10^{-7}$, $K_{a_3} = 10^{-11}$,

13. How many electrons are present in gerade molecular orbital of C_2 .

14. Calculate pH at which $Mg(OH)_2$ begins to precipitate from a solution containing 0.1 M Mg^{2+} ions.

K_{sp} of $Mg(OH)_2 = 10^{-11}$

15. Assuming there no 2s – 2p mixing, the number of paramagnetic species amongst the

following: $H_2, He_2^+, Li_2, Be_2, B_2, C_2, C_2^{2-}, N_2^+, N_2, O_2, O_2^-, S_2$ & F_2 are _____.

SECTION-3 : MATHEMATICS**PART – A**
(Numerical based)

This section contains **15 questions**, numerical based questions, (answer of which maybe positive or negative numbers or decimals).

1. It is given that $\log_6 a + \log_6 b + \log_6 c = 6$, where a, b, c are positive integers that form an increasing geometric sequence and $b - a$ is the square of an integer. Find $a + b + c$.
2. Suppose that x and y are real numbers such that $\log_x 3y = \frac{20}{13}$ and $\log_{3x} y = \frac{2}{3}$. The value of $\log_{3x} 3y$ can be expressed in the form $\frac{a}{b}$ where a and b are positive relatively prime integers. Find $a + b$.
3. Given that $\log_{10} \sin x + \log_{10} \cos x = -1$ and that $\log_{10} (\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, find n .
4. If exactly one root of the equation $4x^2 - 4ax + a^2 - 2a + 2 = 0$ lies in the interval $(0, 2)$ then the least value of a is
5. If the number of integer values that a can take so that the expression $\frac{x^2 + 2x + a}{x^2 + 4x + 3a}$ is capable of taking all real values for all possible real values of x is k then $k + 2 =$
6. If the expression $\sqrt[8]{2207 - \frac{1}{2207 - \frac{1}{2207 - \frac{1}{2207 - \dots}}}}$ can be expressed in the form $\frac{a + b\sqrt{c}}{d}$ where a, b, c, d are not composite integers then value of $a + b + c + d$ equals
7. Normals are drawn to a parabola at points A, B and C on it which are concurrent at $N \equiv (4, 2)$. If $A \equiv (3, 3)$ and $B \equiv (1, -1)$ and the focus S of the parabola be (α, β) then the value of $2\alpha - \beta$ is
8. Given that $x + y = 2$ is a tangent to an ellipse at the point $P(\alpha, \beta)$ on it and its foci are $(2, 3)$ and $(3, 5)$. If e be the eccentricity of the ellipse then find the value of $\frac{\alpha + \beta}{e^2}$
9. An ellipse is known to have its centre at $(3, 4)$ and it touches the x -axis at $(0, 0)$. If slope of the major axis is 1 then find the eccentricity of the ellipse.
10. A tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with centre C meets its director circle at P and Q . Then the product of the slopes of CP and CQ is

11. Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to $2R$ and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then $4e_2^2 - e_1^2 =$
12. If the locus of the middle points of the chords of the parabola $y^2 = 2x$ which touches the circle $x^2 + y^2 - 2x - 4 = 0$ is given by $(y^2 + 1 - x)^2 = \lambda(1 + y^2)$, then the value of λ is equal to
13. If one of the roots of the equation $2000x^6 + 100x^5 + 10x^3 + x - 2 = 0$ is of the form $\frac{m + \sqrt{n}}{r}$ where m is a non-zero integer and r and n are relatively prime natural numbers, then the value of $(m + n + r)$ is
14. The number of integral values of a for which the inequality $3 - |x - a| > x^2$ is satisfied by at least one negative x is
15. Tangent is drawn at any point (x_1, y_1) on the parabola $y^2 = 4ax$. Now tangents are drawn from any point on this tangent to the circle $x^2 + y^2 = a^2$ such that all the chords of contact pass through a fixed point (x_2, y_2) . Then $4\left(\frac{x_1}{x_2}\right) + \left(\frac{y_1}{y_2}\right)^2 + 1$ equals

PERFORMANCE ENHANCEMENT TEST – 6**ANSWERS****SECTION-1 : PHYSICS****PART – A**

- | | | | |
|-----|----------------------------|-----|-------------------------------|
| 1. | 1.41 | 2. | 1 |
| 3. | 6.50 | 4. | 1.13 |
| 5. | 4.24 | 6. | 14.67 (range: 14.60 to 14.70) |
| 7. | 3.87 (range: 3.86 to 3.88) | 8. | 0.12 |
| 9. | 0.25 | 10. | 1 |
| 11. | 6.25 | 12. | 28.28 |
| 13. | 23.75 | 14. | 0.15 |
| 15. | 1.50 | | |

SECTION – 2 : CHEMISTRY**PART – A**

- | | | | |
|-----|-----|-----|------|
| 1. | 8 | 2. | 3 |
| 3. | 5 | 4. | 2.75 |
| 5. | 6 | 6. | 1 |
| 7. | 3 | 8. | 8 |
| 9. | 9.7 | 10. | 9 |
| 11. | 2 | 12. | 7 |
| 13. | 4 | 14. | 9 |
| 15. | 6 | | |

SECTION – 3 : MATHEMATICS**PART – A**

- | | | | |
|-----|---------------------------------|-----|----------------------------------|
| 1. | 111 | 2. | 199 |
| 3. | 12 | 4. | 2.35 (range from (2.3 to 2.4) |
| 5. | 2 | 6. | 11 |
| 7. | 3 | 8. | 16.4 |
| 9. | 0.928 (range from 0.92 to 0.93) | 10. | -0.44 (range from (-0.4 to -0.5) |
| 11. | 6 | 12. | 5 |
| 13. | 200 | 14. | 6 |
| 15. | 1 | | |